

Autonom8: Helping businesses reduce operating costs and grow profit with Google Cloud

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About Autonom8

With Google Cloud, Autonom8 can run on a low-code platform that accelerates and streamlines customer journeys in a scalable, reliable, cost-effective infrastructure, with advanced optical

Headquartered in the United States and India, [Autonom8](https://autonom8.com/) (<https://autonom8.com/>) has built a low-code SaaS platform that allows businesses to digitize customer-facing workflows. The business aims to help clients reduce costs and improve interactions with their customers through automation and enablement of customer journeys.

Industries: Technology

recognition to enable intelligent document processing.

Location: United States and India

Google Cloud results

- Increased margins by up to 30% by switching from a home-grown OCR system to Cloud Vision AI
- Enables one DevOps team member to manage up to 30 customers
- Provides real-time information about customer journeys to enable businesses to respond quickly and accurately
- Ensures use of its platform with

Red
open
cost
20%
local
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archi
through

(<https://workspace.google.com/>), Google

Google Cloud (<https://cloud.google.com/>)
s/drive/

Cloud SQL (<https://cloud.google.com/sql/>)
, Gmail

Vision AI (<https://cloud.google.com/vision/>) mail
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Kubernetes Engine
(<https://cloud.google.com/kubernetes-engine/>)
s/)

Google Workspace
(<https://workspace.google.com/>)

Google Drive
(<https://workspace.google.com/products/drive/>)

Gmail
(<https://workspace.google.com/products/gmail/>)

Sheets
(<https://workspace.google.com/products/sheets/>)

**containerization
in customers'
private data
centers**

Just as cars are evolving to become autonomous, smart and self-driving, enterprises can gain self-awareness, an ability to learn and an ability to adapt. This is the value proposition put forward by Autonom8, an India- and United States-based enterprise workflow management software business. "We provide a low-code, high-intelligence customer journey automation SaaS platform," explains Ranjit Padmanabhan, Co-founder and Chief Technology Officer (CTO), Autonom8.

The Autonom8 platform includes components, such as A8Studio, a drag and drop location from which clients can create customer journeys, a chat platform that enables clients to create chatbots, and an analytics module. "With our platform and services, businesses can reduce costs and improve interactions with their customers by applying automation to accelerate and provide better customer journeys," adds Padmanabhan.

Demand for Autonom8 is being driven by the changing customer demands of enterprises, including the expectation to interact with them over multiple channels, and the rising cost of building software with experienced developers. These trends place enterprises under growing pressure to increase the productivity of the people they do have, particularly those who are less technically inclined.

In addition, changing consumer habits, regulations and the emergence of new technologies mean customer journeys cannot remain static and need to evolve.



Autonom8 Team

Developing a microservices-based SaaS platform

From the start, Autonom8 planned to deliver a SaaS platform and initially deployed on a multinational cloud service, chosen due to the team's familiarity with its products and the availability of credits.

However, the company's decision to opt for a microservices architecture that enables individual services to scale independently while running in a containerized environment, demanded high-quality container orchestration. To optimize cost, scalability and performance, Autonom8 began evaluating

Google Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine>) (GKE).

The business then completed a side-by-side comparison between Google Cloud and its incumbent provider of compute, storage and other services. Google Cloud fared favorably, with Vision AI (<https://cloud.google.com/vision>) in particular providing powerful machine learning and optical character recognition (OCR) functionality, supporting a key use case for Autonom8.

In addition, many of Autonom8's clients at the time are financial institutions in India, and legally required to retain data within the country's borders. Google Cloud's global network and local presence means the business could fulfill this requirement easily.

"We decided to evaluate Google Cloud, particularly GKE, from two perspectives. One, from a security perspective, as we sell to banks that audit our platform, and two, as a failover between regions because downtime costs money. We found it a compelling solution," says Padmanabhan.

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A seamless move to Google Cloud

Autonom8 began deploying on Google Cloud in 2018, with its architecture comprising storage, compute, serverless, container management and orchestration, and Vision AI. "We looked at our scripting with the previous provider, and using the Google Cloud documentation available online, educated ourselves over a few weeks before moving pieces of our architecture step by step to Google Cloud," says Padmanabhan. "We did not run into any

major issues. It was pretty simple, with our experienced engineers training others in the product."

According to the CTO, the business had two options when moving to Google Cloud. Autonom8 could either install raw virtual machines and effectively create its own virtualized data center, or rely on managed services for functions such as memory store, registration and authentication to save time and resources over the long term. Autonom8 opted for the latter and has transitioned fully to Google Cloud, with the number of cloud products and services in its architecture rising from five to about 15. While each product and service performs a key role in the delivery of Autonom8's products and services, Padmanabhan nominates GKE, Vision AI and Cloud SQL (<https://cloud.google.com/sql>) as providing the greatest value to the business.

Scalability, real-time monitoring and intelligent document processing at low cost

With GKE, the business can now scale the nodes or containers specific to each microservice in the event traffic to a particular client surges, due to a rebate or promotion. "Through the combination of the architecture and localized scalability we achieve with GKE, we are reducing our operating costs by up to 20%," says Padmanabhan.

Running an open source TimescaleDB on Postgres in Cloud SQL enables Autonom8 to give its clients the ability to monitor customer journey information

in real time. An example of a journey is applying for a bank loan. The customer must take steps including providing income, tax and other financial details that the bank then appraises to help make a decision on the application. "The moment someone applies for a loan, for example, a bank knows about it and can monitor for fraud, bottlenecks, or other abnormalities, and immediately route to a remediation workflow," explains Padmanabhan. "Cloud SQL enables us to maintain transactional logging and provide real-time data to our dashboards."

After evaluating alternative services, including developing a home-grown OCR engine, the business turned to Cloud Vision AI to manage the intelligent document processing that comprises much of its transactional volume. "Vision AI is significantly better than the alternatives and the cost of maintaining our version did not make sense, because Google Cloud continues to make improvements over time that enable us to deliver more and more accurate results to our customers," says Padmanabhan. "Switching from our home-grown service to Vision AI has enabled us to increase our profit margins by up to 30%."

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Supporting client demands and improving developer efficiency

Google Cloud also enables Autonom8 to meet the demands of businesses that want to run its platform within their own private data centers. "We can undertake the build within Google Cloud and ship our containers to compatible hosts within those clients' data centers," explains Padmanabhan. "With our previous provider, we could create containers, but these would not run properly within those data centers."

Furthermore, Google Cloud documentation and online resources help Autonom8 reduce the training needed for new developers to become productive,

with the Google Cloud learning curve taking up just 10% of the overall onboarding cycle.

The organization spends the equivalent of just 3% of its overall annual revenue on DevOps, measured as DevOps Utility Ratio, while the cloud cost of revenue is about USD 1 for every USD 6 in annual recurring revenue, measured as Cloud Utility Ratio. "These two metrics are about what we can do with the people we have," explains Padmanabhan. "Our current ratio implies that one DevOps person can handle approximately 30 customers. This is made possible by the tools we have, and the comprehensive support from Google Cloud in terms of security patches, intelligent alerts, resource overloading, and more."

Google Cloud also provides the flexibility for Autonom8 to accommodate the varying service levels required by individual customers based on factors, such as the impact of downtime, as the business can failover seamlessly between regions to mitigate the impact of any issues that may occur.

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Integrating Google Workspace with Autonom8 to deliver new capabilities

Autonom8 relies on Google Workspace (<https://workspace.google.com/>) for communication, collaboration and other workplace productivity requirements, growing its footprint from Gmail (https://workspace.google.com/intl/en_au/products/gmail/) when the employee population was four or five, to a

range of products including Sheets

(https://workspace.google.com/intl/en_au/products/sheets/)

and Drive

(https://workspace.google.com/intl/en_au/products/drive/)

as the business grew. "It became natural to use the capabilities in Google Workspace as we matured," says Padmanabhan. "One of the most interesting capabilities was our ability to integrate Google Workspace into our platform. For example, when someone is running a workflow, they can add data from a Sheet. We've added Google Workspace authentication capabilities into our products as well."

"Everyone is using shared links to Drive and I really like the granular permissions structure," he adds. "I can open up folders to clients while keeping an internal space within the business to ensure security and privacy."

With Google Cloud, Autonom8 is now poised to continue growing its business and adding new features and capabilities for clients. "We are extremely excited at the opportunity to step up our offering to clients with Google Cloud," concludes Padmanabhan.